SPHEREx FITS Structure & Data Analysis

TEAM REPORT | Task S1-13: IMAGE, FLAGS, and VARIANCE

Dataset	SPHEREx Data Release 1 (DR1)
Observation ID	2025W19_1B_0116_1
Selected product	level2_2025W19_1B_0116_1D3_spx_l2b-v27-2026-197.fits
Environment	Google Colab | Python | Astropy | NumPy | Matplotlib

Objective

Inspect a real SPHEREx FITS Level 2 product, identify its HDU structure, extract IMAGE, FLAGS, and VARIANCE, and document their data types, dimensions, numerical properties, and visual appearance. This establishes a data-understanding baseline for later image preprocessing and temporal comparison.

Data source and environment

The data were located through the NASA/IPAC Infrared Science Archive (IRSA) SPHEREx Spectral Image Search. The selected observation was identified as 2025W19_1B_0116_1, with product filename level2_2025W19_1B_0116_1D3_spx_l2b-v27-2026-197.fits. The file was downloaded and inspected in Google Colab using Python, Astropy, NumPy, and Matplotlib.

FITS structure

The file contains seven HDUs. IMAGE, FLAGS, VARIANCE, and ZODI are image-like extensions; EPSF and WCS-WAVE are binary tables. The three main analysis arrays IMAGE, FLAGS, and VARIANCE share the same 2040 × 2040 pixel grid. FLAGS is stored as a compressed image HDU (CompImageHDU).

HDU	Extension	HDU type	Dimensions / format
0	PRIMARY	PrimaryHDU	No image array; 5 header cards
1	IMAGE	ImageHDU	2040 × 2040; float32; 276 cards
2	FLAGS	CompImageHDU	2040 × 2040; int32; 185 cards
3	VARIANCE	ImageHDU	2040 × 2040; float32; 119 cards
4	ZODI	ImageHDU	2040 × 2040; float32; 122 cards
5	EPSF	BinTableHDU	451 rows × 11 columns
6	WCS-WAVE	BinTableHDU	1 row × 3 columns

Numerical summary

Property	IMAGE	FLAGS	VARIANCE
Shape	(2040, 2040)	(2040, 2040)	(2040, 2040)
Data type	float32 (>f4)	int32	float32 (>f4)
Minimum	-9997.668	0	1.5366577 × 10■■
Maximum	48396.996	87 unique values	2.1015505 × 10■
Mean	0.34205776	—	0.71973157
Median	Not calculated	—	0.00023914097
NaN pixels	233	Not reported	233

IMAGE findings

IMAGE is a 2040 × 2040 float32 array. Its observed minimum was −9997.668, maximum 48396.996, mean 0.34205776, and 233 NaN pixels. Negative values should not automatically be treated as invalid: calibrated astronomical measurements may include negative background-subtracted values. The extreme minimum should be interpreted using product documentation and quality information.

FLAGS findings

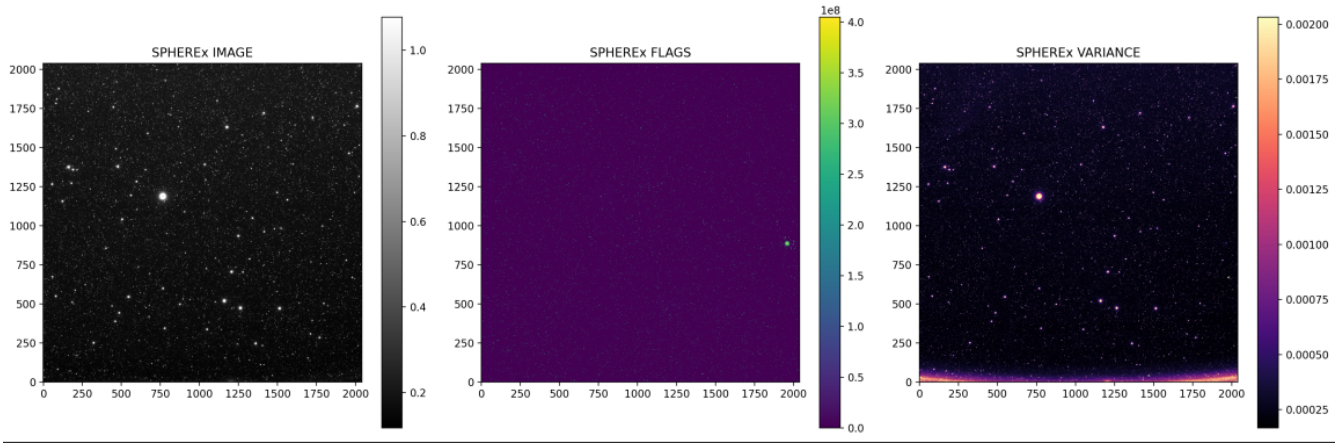
FLAGS is a 2040 × 2040 int32 mask array with 87 distinct integer values. A total of 1,873,676 pixels had a value of zero. The header identifies the extension as a mask (EXTTYPE='MASK') and provides WCS-related coordinate metadata. No MP_* flag-definition keywords were found in the inspected FLAGS or IMAGE headers.

VARIANCE findings

VARIANCE is a 2040 × 2040 float32 array. Its minimum was 1.5366577 × 10■■, maximum 2.1015505 × 10■, mean 0.71973157, median 0.00023914097, and 233 NaN pixels. No zero or negative variance pixels were found. Percentiles were: 25th 0.000211111101, 50th 0.00023914097, 75th 0.00027788198, 90th 0.00034761085, 95th 0.00053986197, 99th 0.002031076, and 99.9th 0.05623471.

Visual inspection

The combined visualization shows IMAGE with a grayscale percentile stretch, FLAGS with a color map, and VARIANCE with percentile-limited display scaling. The FLAGS color scale represents integer mask values; it is not a direct good/bad quality scale. The bright structure near the lower edge of the VARIANCE image is visible, but its cause has not been established in this analysis.



Exploratory FLAGS bit counts

Exploratory bit label*	Pixels with bit set
TRANSIENT	19,713
OVERFLOW	829
SUR_ERROR	13,527
NONFUNC	12,153
DICHROIC	109,713
MISSING_DATA	0
HOT	1
COLD	1,766
FULLSAMPLE	272
PHANMISS	0
NONLINEAR	0
PERSIST	1,905
OUTLIER	77,002
CROSSTALK	0
SOURCE	2,129,157
GHOST	0

*The names shown are provisional labels used for exploratory bit tests, not a verified authoritative mapping for this exact DR1 product. No MP_* definitions were found in the inspected IMAGE or FLAGS headers. Pixels may have multiple bits set, so counts overlap. Do not reject pixels using this table until the exact DR1 bit definitions and handling guidance are verified.

Scope and limitations

This task focused on FITS structure and inspection of IMAGE, FLAGS, and VARIANCE. No image alignment, image subtraction, source detection, or final quality-mask rejection was performed. Bit counts are counts of pixels for which each selected bit test was true; multiple bits may be set on one pixel, so counts overlap. The bit-name mapping used in exploratory analysis is provisional until checked against the exact SPHEREx DR1 product documentation. These counts must not be used as a definitive pixel-rejection policy.

Conclusions and handoff

The FITS file was successfully downloaded, opened, and inspected. The main arrays were extracted, visualized, and numerically summarized. Their matching dimensions provide a basis for later pixel-by-pixel analysis. Before constructing a Quality Mask, the team should verify the exact flag-bit definitions and recommended handling rules for this DR1 product.

Team handoff checklist

1. Executed Jupyter Notebook (.ipynb) with code and outputs.
2. Combined visualization PNG (IMAGE, FLAGS, VARIANCE).
3. This report PDF and editable DOCX.
4. Keep the original FITS file unchanged; use a separate derived copy for later processing.